



Technische
Universität
Braunschweig

**Decision
Support**
Institut für Wirtschaftsinformatik



Data Driven Decision Making

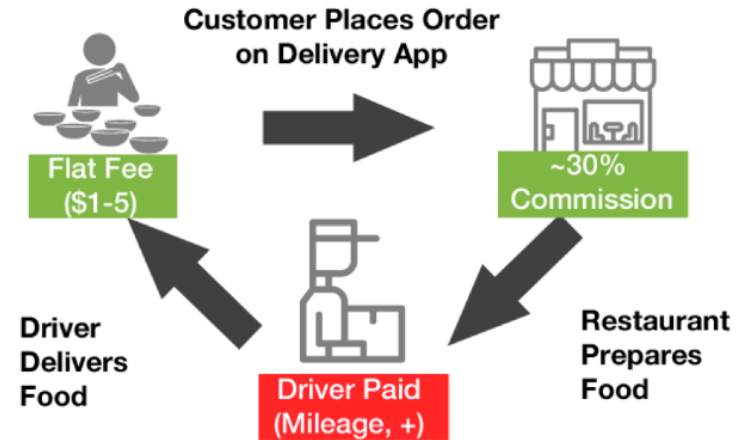
Lecture 11 – Platform-based Food Delivery

Content

- Platform-based Food Delivery
- The Eater's Perspective
- Uber Eat's Perspective
- The Delivery Perspective
- The Restaurant's Perspective

Platform-based food delivery

- Platform-based business models act in a gig economy with volatile business partners
- Uber Eats introduces the three sided-market place
- Revenue mainly from restaurants (+user flat fee)
- Payments are made to delivery drivers
- Platform is not involved in physical transaction

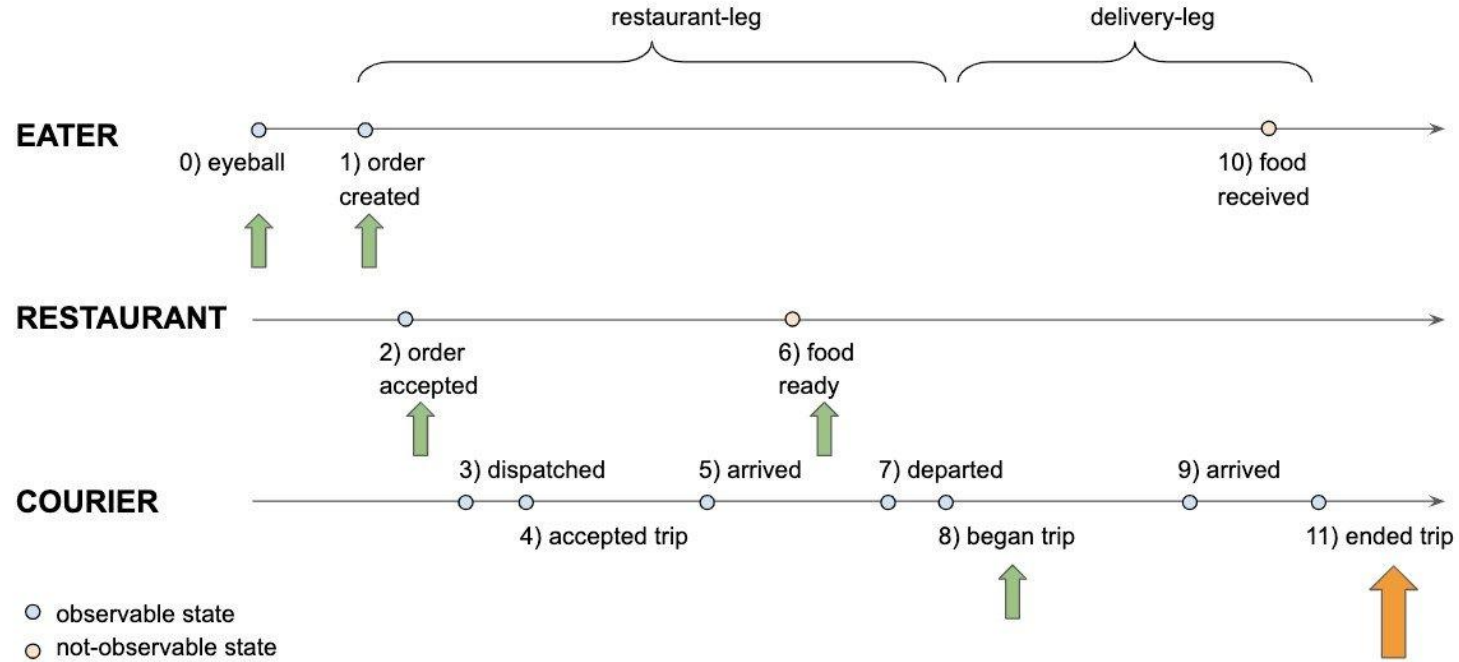


<https://www.intelivita.com/blog/food-delivery-business-model/>

Properties

- The platform business model is applicable in different regions and scales in any order
- Decisions needed for the transactional control are taken in a fully automated fashion
- Partners in the three-sided market place occur and vanish at random
- The credibility of partners is maintained by monitoring, data analysis and ranking
- Data driven decision making is the key feature for running such a platform

Interaction of partner



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

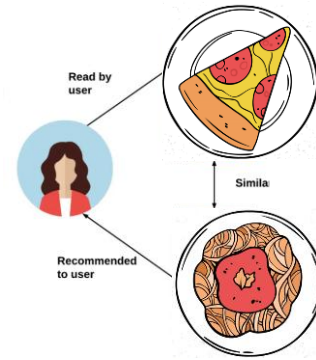
Content

- Platform-based Food Delivery
- **The Eater's Perspective**
- Uber Eat's Perspective
- The Delivery Perspective
- The Restaurant's Perspective

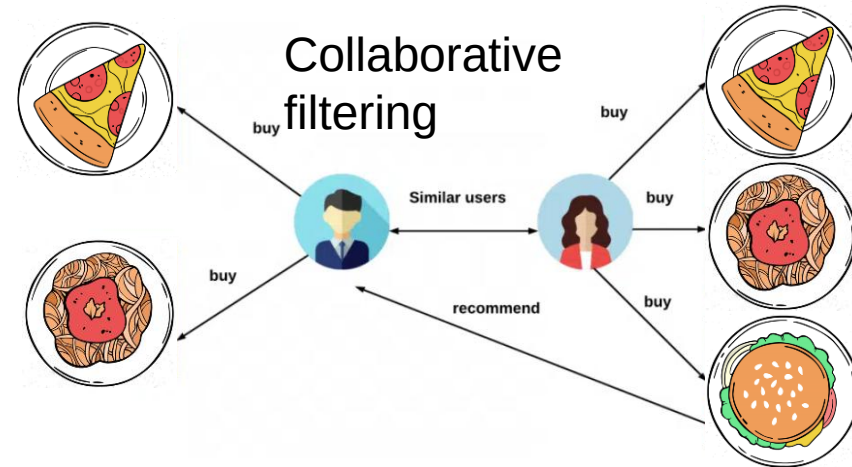
Stages in generation of food recommendation

1. Non-personalized recommendation focus on very few restaurants only
 2. Content-based recommendations have led to a meal over-specialization
 3. Collaborative filtering is used for personalized recommendations
- 3. requires the clustering of eaters in accordance to personal attributes (not considered here any further)

Content based



Collaborative filtering



<https://thingsolver.com/blog/introduction-to-recommender-systems/>

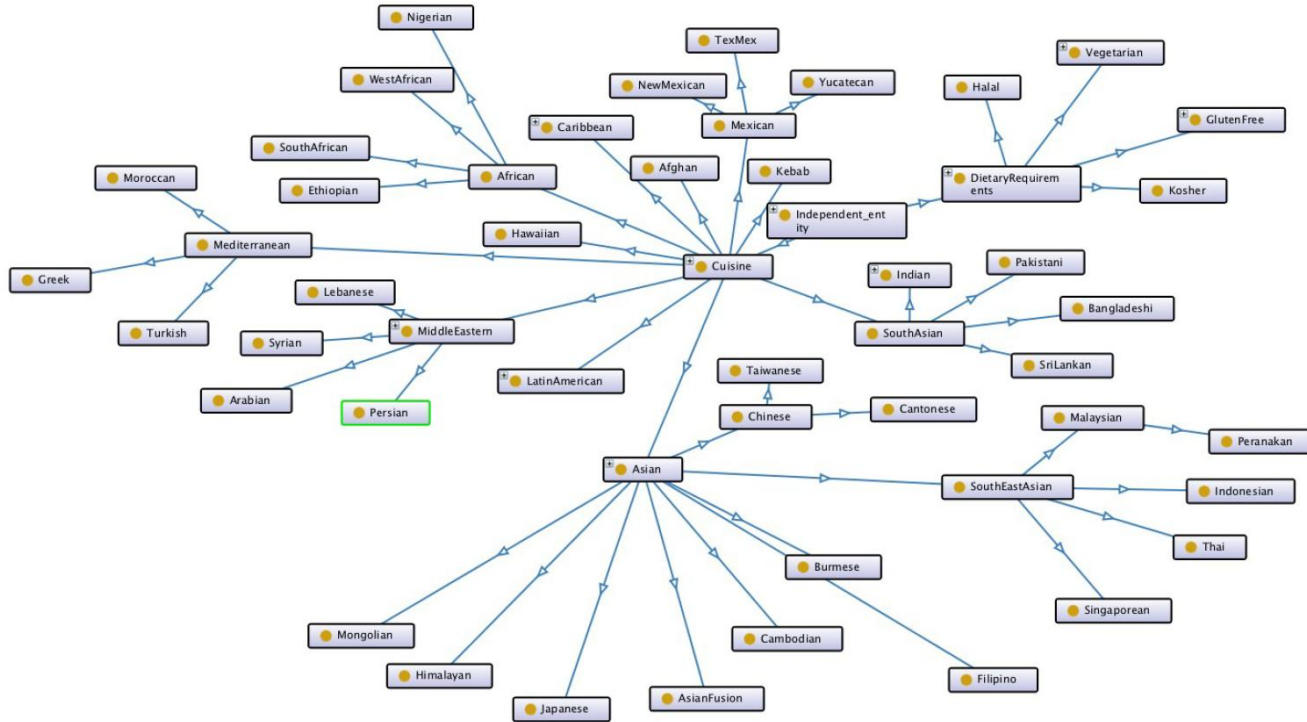
The eater's perspective (non-personalized recommendation)

- The eater has got a preferred cuisine
- She prefers food from high ranked restaurants
- Fast delivery is of utmost importance
- Reasonable pricing is desirable
- However, restaurant choices may be too similar
- There should be some range of food offered to the eater
- Diverse food offerings maintain eater's retention

Ken Ken Ramen	20-30 MIN
Ramen-Japanese-\$\$	
Genki Ramen	25-35 MIN
Seafood-Japanese-\$\$	
Jika Ramen & Sushi	30-40 MIN
Sushi-Japanese-\$	
The Ramen Bar	35-45 MIN
Ramen-Japanese-Noodles-\$\$	
Uchiwa Ramen	40-50 MIN
Ramen-Japanese-\$\$	
Ramen Izakaya	35-45 MIN
Sushi-Japanese-\$\$	
Ramen Doraku	30-40 MIN
Japanese-Ramen-\$\$	

<https://www.uber.com/en-DE/blog/uber-eats-trip-optimization/>

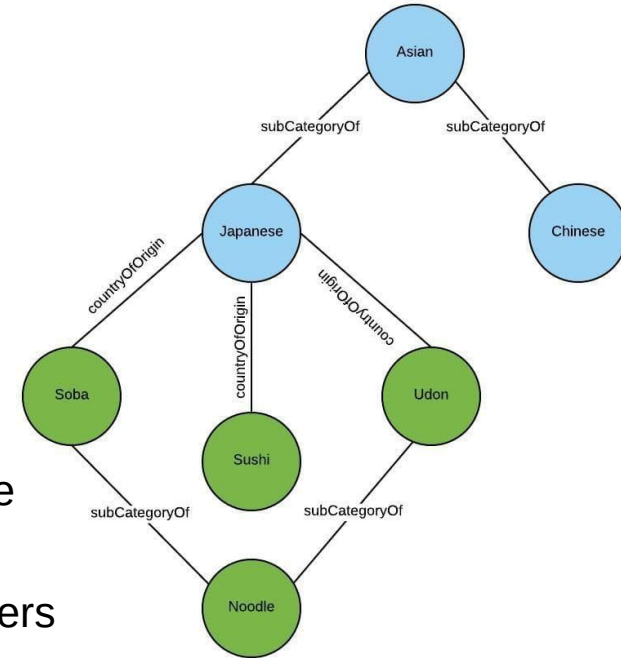
Food ontology graph (content based recommendation)



<https://www.uber.com/en-DE/blog/uber-eats-query-understanding/>

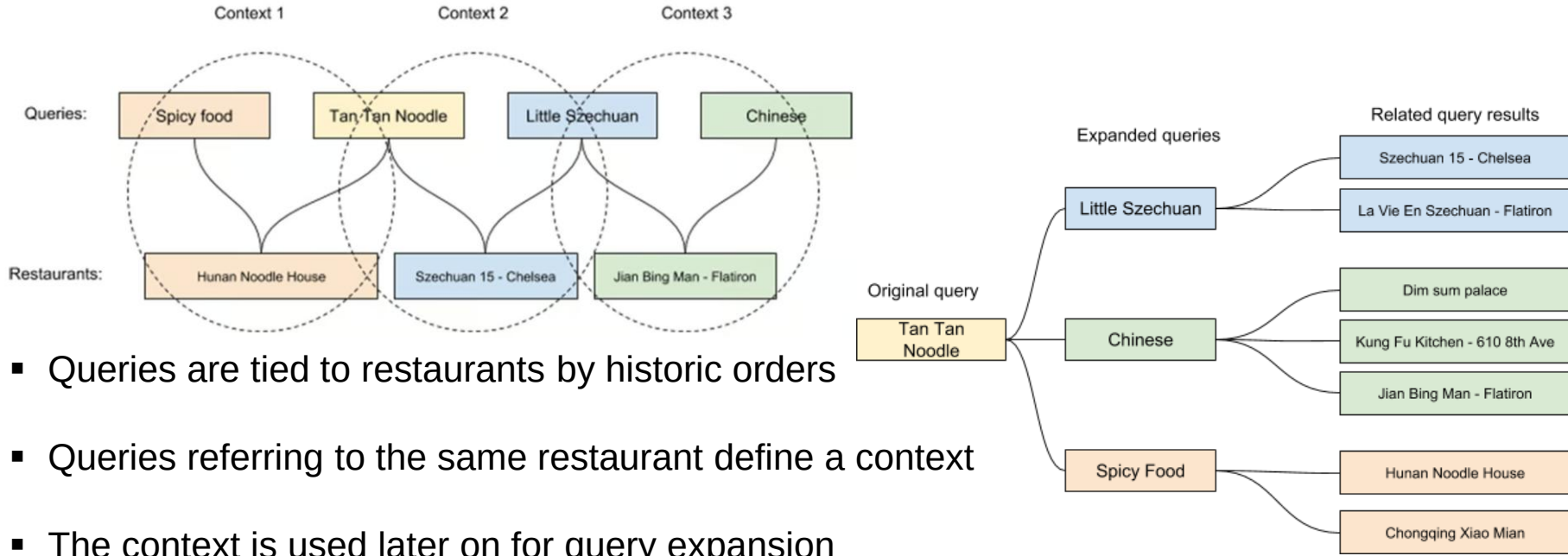
Content based diversification of meal offerings

- Ontology subgraph links eater query with alternative cuisine
- Having searched for Soba, also Udon may be offered
- Broaden the scope may also offer Sushi, or Chinese dishes
- Ontologies are built offline by incorporating expert knowledge
- The online counterpart for diversification is learning from orders



<https://www.uber.com/en-DE/blog/uber-eats-query-understanding/>

Online diversification (collaborative filtering)

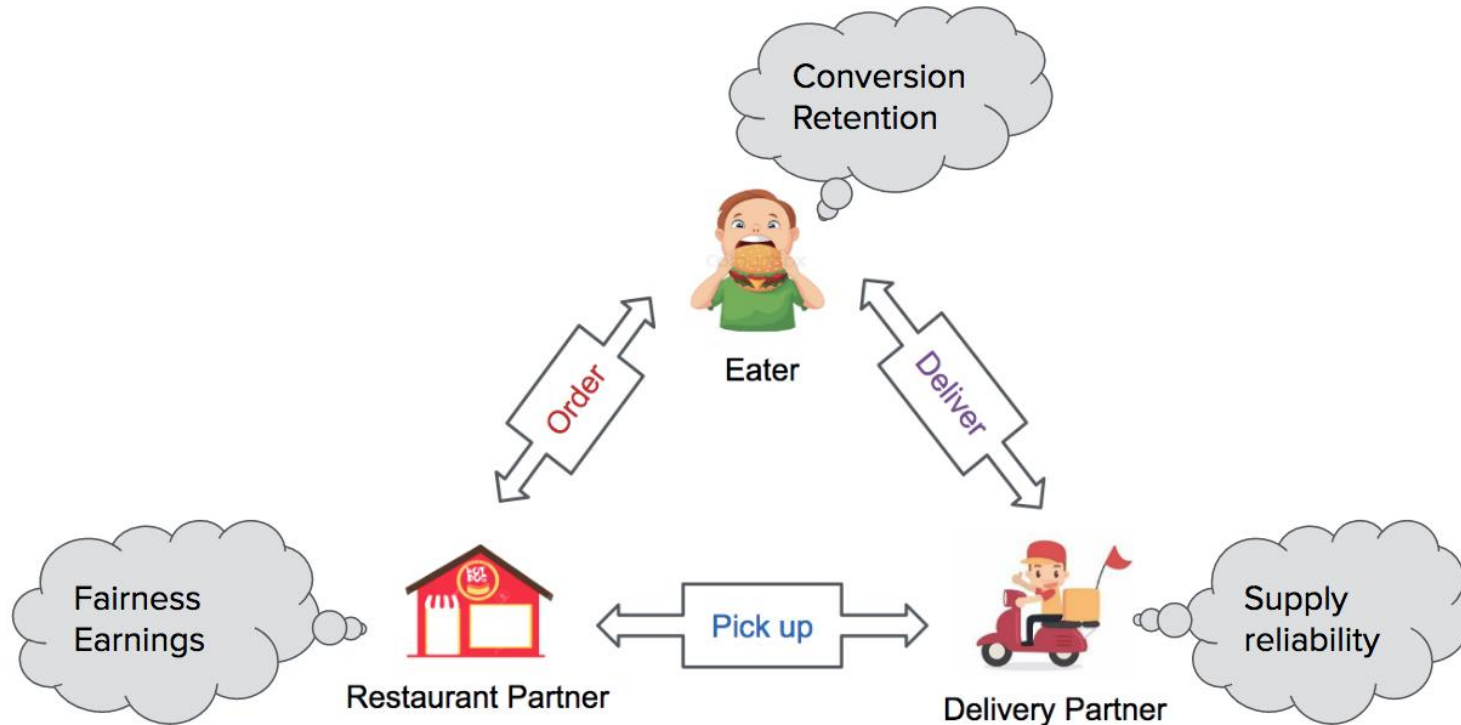


<https://www.uber.com/en-DE/blog/uber-eats-query-understanding/>

Content

- Platform-based Food Delivery
- The Eater's Perspective
- **Uber Eat's Perspective**
- The Delivery Perspective
- The Restaurant's Perspective

Controlling the marketplace

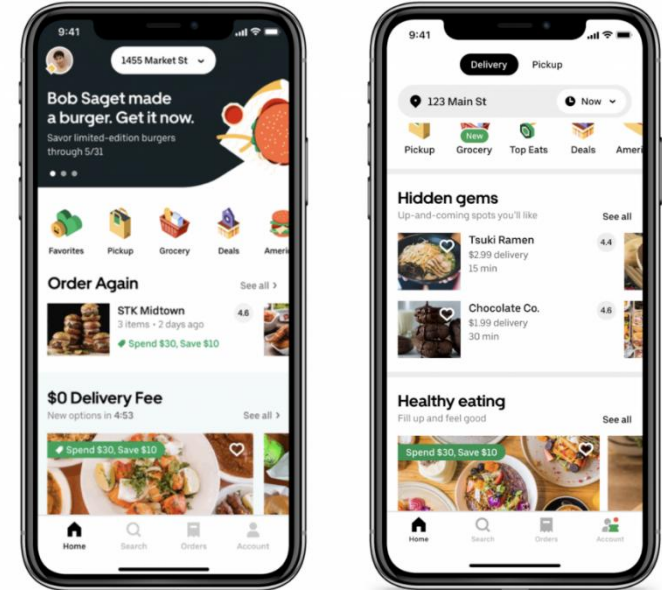


<https://www.uber.com/en-DE/blog/uber-eats-trip-optimization/>

Ranking of recommendations

- Approximately 80% of order are picked from recommended meals shown in app
- Recommendations should be generated in a way that aligns with partner's objectives

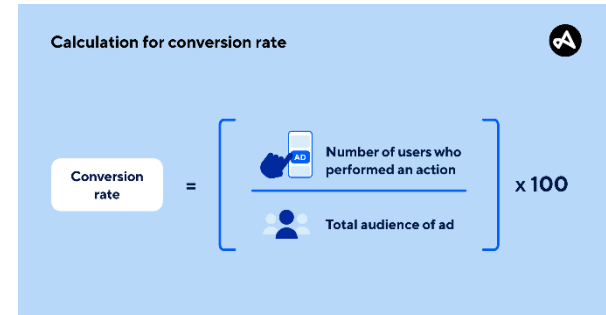
Partner	Objective
Eater	Conversion
	Retention
Restaurant	Fairness
	Earnings
Driver	Supply reliability



<https://www.uber.com/newsroom/arriving-now-the-new-uber-eats/>

Incorporate multiple (contradicting) objectives

- Gross booking value is defined as the full amount of payments processed on the platform. It is achieved by placing pricy orders at popular restaurants and addresses restaurant earnings
- Marketplace fairness is in the interest of new or less popular restaurants. Earnings from such restaurants should be generated in order to prevent such restaurants from leaving the platform
- A conversion rate records the percentage of users who have completed a desired action. Here, it defines a user interaction that results in a meal order. This strengthens eater retention
- Gross booking value conflicts with marketplace fairness
- Gross booking value may conflict with conversion rate
- Conversion rate and marketplace fairness are independent of each other or may be even complementary



<https://www.adjust.com/glossary/conversion-rate/>

Modeling (1)

Model decides about the priority of recommending (reasonable) restaurants to eater

x_{ij} = decision variable that represents the probability of recommending restaurant j to eater i

I_{ij} = indicator of restaurant j being recommended by eater i

O_{ij} = indicator of eater i ordering from restaurant j

G_{ij} = dollar amount of the order between eater i and restaurant j

Conversion rate $T(X) = E[\sum_i \sum_j I_{ij} O_{ij}] = \sum_i \sum_j x_{ij} p_{ij}$ with p_{ij} predicted conversion rate

Gross booking $G(X) = E[\sum_i \sum_j I_{ij} O_{ij} G_{ij}] = \sum_i \sum_j x_{ij} p_{ij} g_{ij}$ with g_{ij} expected booking value

Both, $p_{ij} = E[O_{ij}]$ and $g_{ij} = E[G_{ij}]$ are personalized machine learning models that use features from eaters, restaurants and real-time contexts.

Modeling (2)

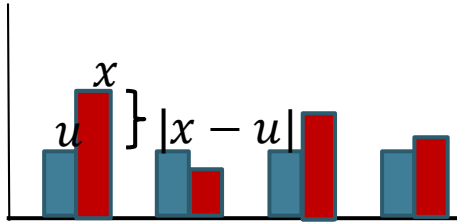
$U = (u_{ij})$ equal probability for all restaurants j to be recommended to eater i with $\sum_j u_{ij} = 1, \forall i$

X^* = probabilities of optimal solution that can achieve best possible conversion rate $T(X^*)$

Parameter $(0 < \alpha < 1)$ for required conversion rate and λ for quadratic fairness penalty

Maximize gross booking value

$$\begin{aligned} \max_X \quad & G(X) - \frac{1}{2} \lambda (X - U)^2 \\ \text{s.t.} \quad & T(X) \geq \alpha T(X^*) \end{aligned}$$



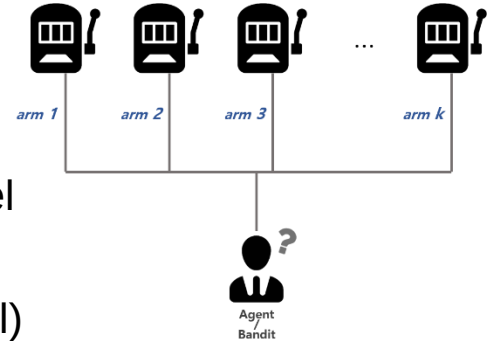
$$\begin{aligned} \sum_j x_{ij} &= 1, \forall i \\ x_{ij} &\geq 0, \forall i, j \end{aligned}$$

Note, that $|x_{ij} - u_{ij}|$ penalizes x_{ij} deviating from u_{ij} twice, therefore penalty is multiplied by $\frac{1}{2}$

As a result, gross bookings are favored that do not need promotion $\rightarrow$ marketplace fairness

Modeling (3)

- Parameter λ penalizes gross booking value that results from restaurant recommendations
 - A small λ favors recommended restaurants and drives out low performing restaurants
 - A high λ enforces that gross booking value is generated from low performers only
- Parameter α enforces a high conversion rate.
 - Setting $\alpha \approx 1$ discharges promising solutions
 - Setting $\alpha \approx 0$ allows for a poor conversion rate
- Both parameters have to be tuned online
- A multi-armed bandit maintains different parameter sets in parallel
- The framework controls whether to perform
 - Exploitation (apply parameter sets that have shown to work well)
 - Exploration (change parameter values in the hope of further improvement)

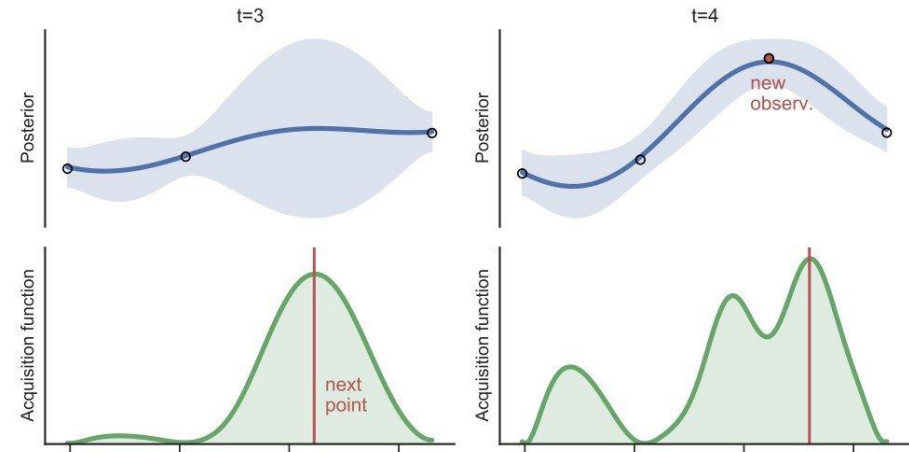


<https://yjo.tistory.com/38>

Modeling (4)

- Exploration by trying a new parameter set is costly in terms of time and profit (loss)
- The number of parameter sets to try is very limited → Bayesian optimization applies

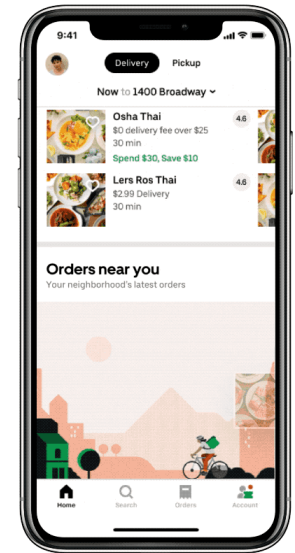
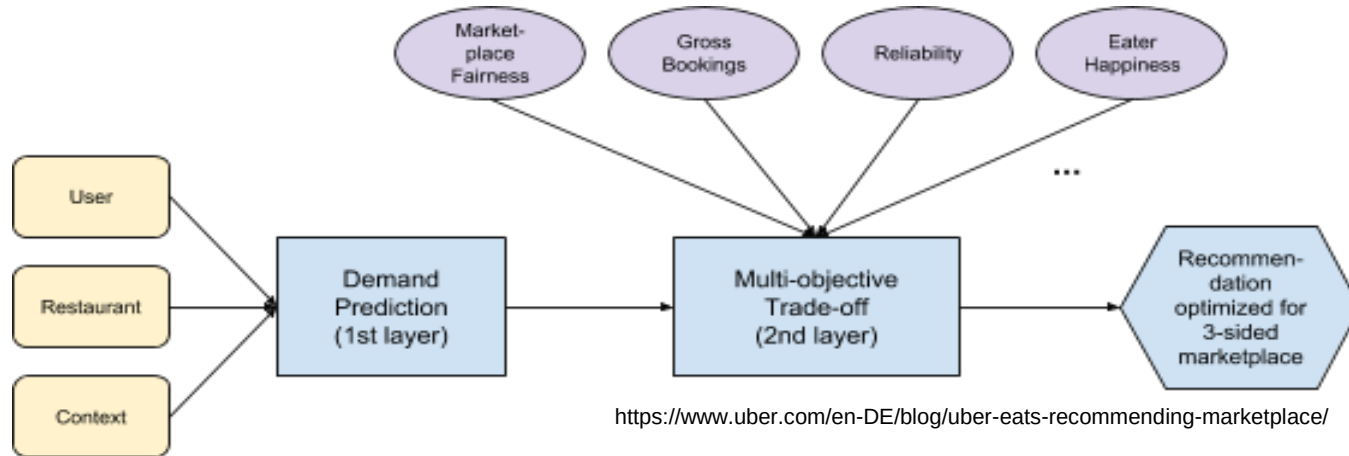
- At $t=3$ three sets have been tested
- For these sets we've got confidence
- An acquisition function suggests new trial
- At $t=4$ the confidence has improved because of the new observation
- A modified acquisition function suggests a new trial at its maximum



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Prediction and optimization pipeline

- Data analysis predicts demand (fulfilment) in the next period
- Multi-objective optimization identifies appropriate restaurants/meals
- Eater's eyeball triggers order that maximizes conversion rate

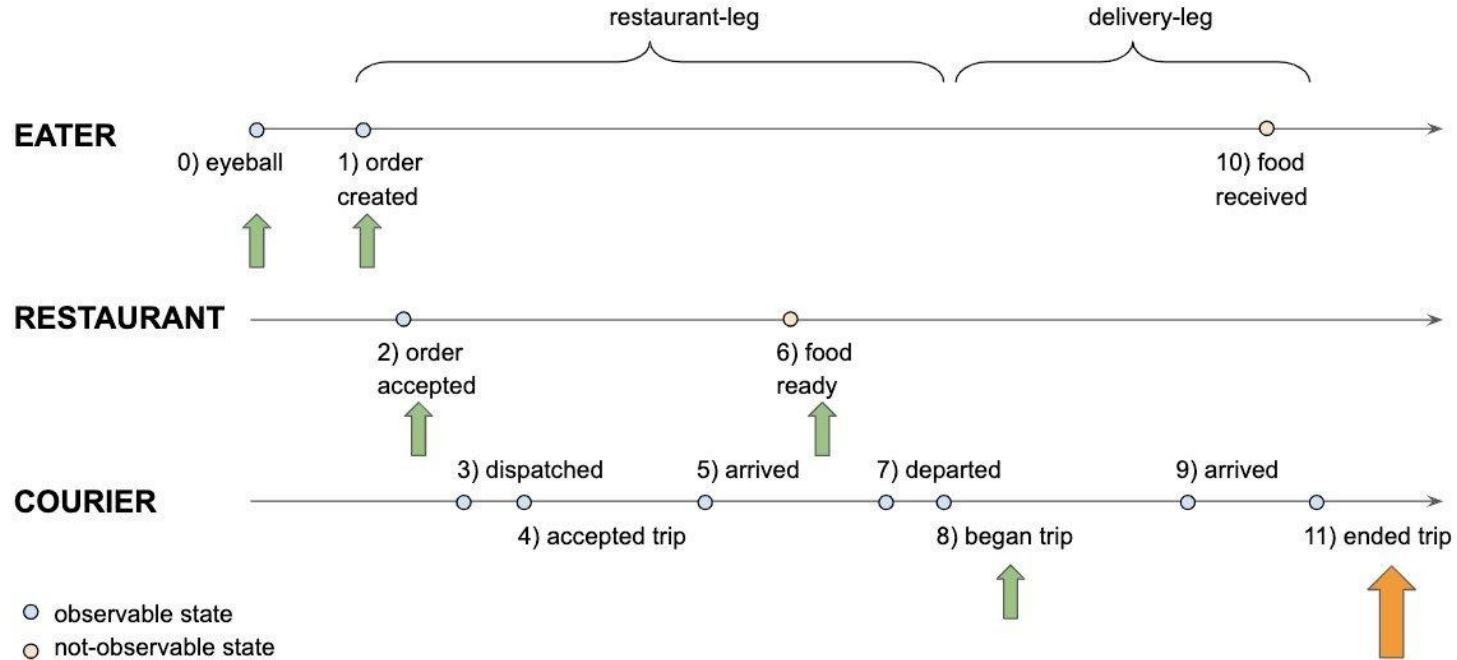


<https://www.uber.com/newsroom/arriving-now-the-new-uber-eats/>

Content

- Platform-based Food Delivery
- The Eater's Perspective
- Uber Eat's Perspective
- **The Delivery Perspective**
- The Restaurant's Perspective

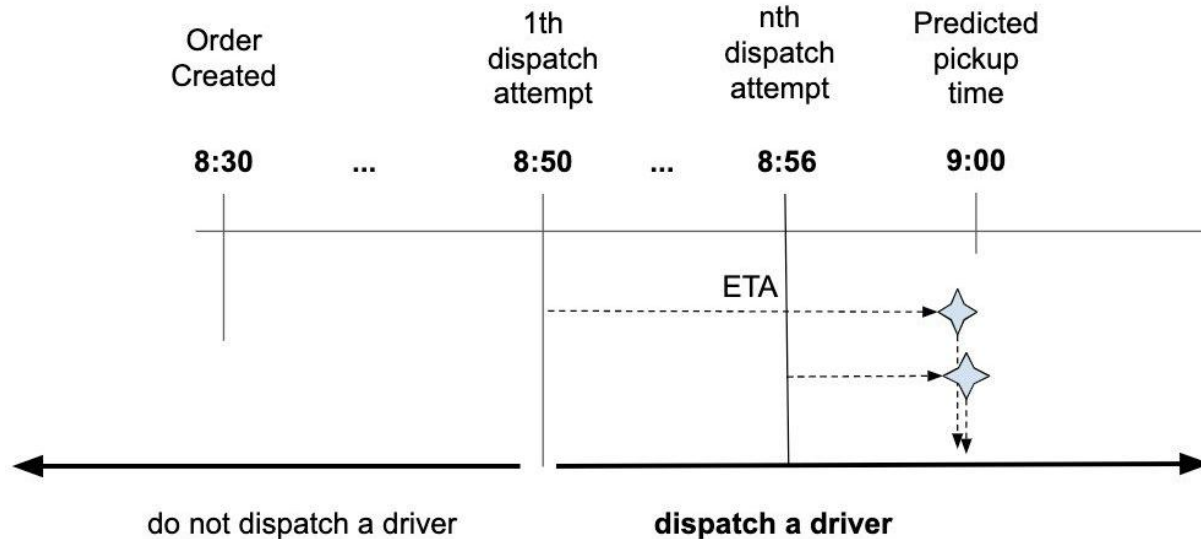
Interaction of partners



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Dispatching a driver

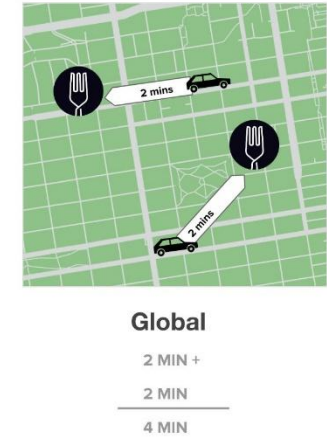
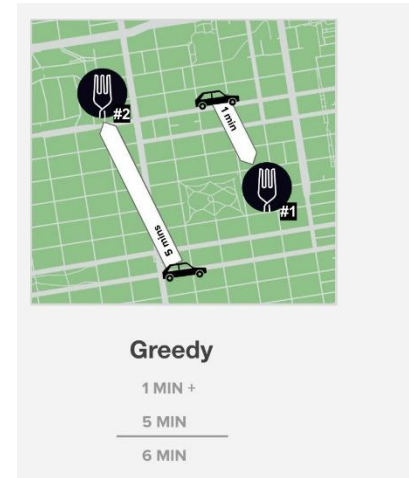
- Given a (30min) meal preparation time, dispatch of drivers starts 10min before ETA
- Several dispatch attempts until successful trip acceptance by one of the driver pool
- Drivers may be ranked and (pre-)selected prior to dispatch attempts



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Driver selection

- Distance to the restaurant
- Second order in vicinity
- Reliability of the driver
- Working time restriction
- Number of recent deliveries
- Weekly payout so far
- Batched opportunities:



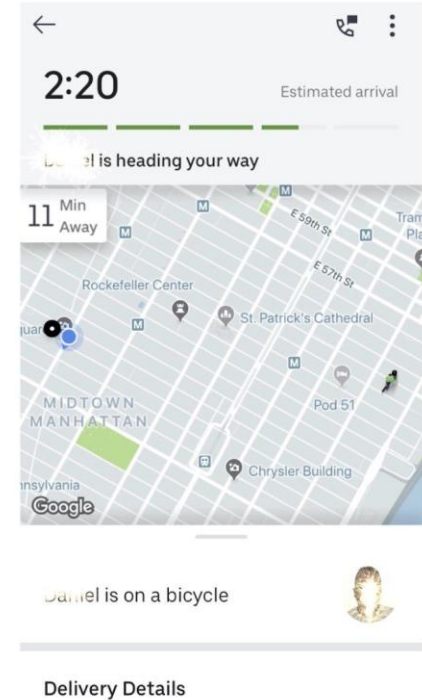
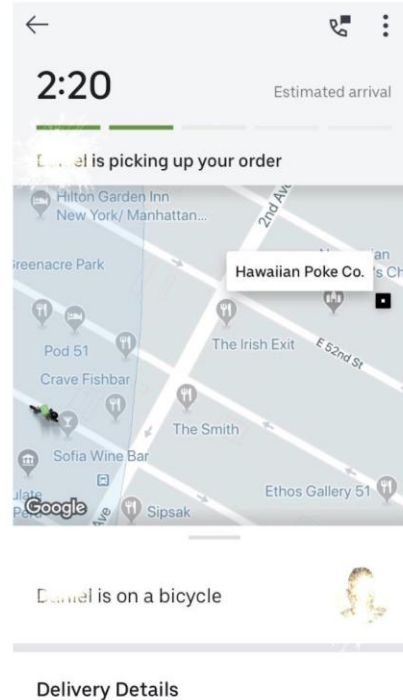
Drivers are paired with orders based on the quickest option for all orders, rather than assigning one driver to one order with a very fast delivery time and another driver to another order with a much slower delivery time

→ Ensuring a roughly comparable waiting time.

<https://www.withpara.com/blog/how-does-uber-eats-assign-drivers>

State of order is communicated to the eater

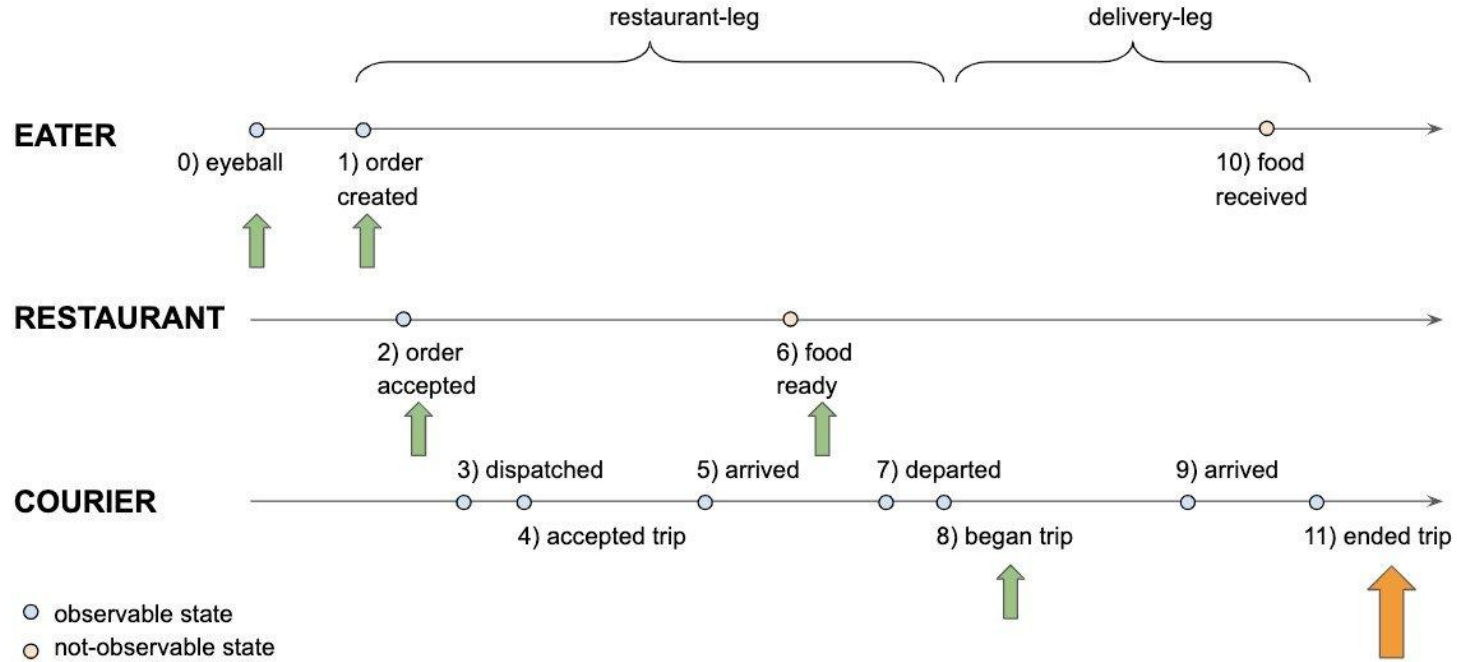
- The driver is monitored
- The position of the driver is communicated to the eater
- The predicted time of delivery is updated
- Drivers are ranked by the platform
 - By the speed of delivery
 - By the reliability of activities



Content

- Platform-based Food Delivery
- The Eater's Perspective
- Uber Eat's Perspective
- The Delivery Perspective
- **The Restaurant's Perspective**

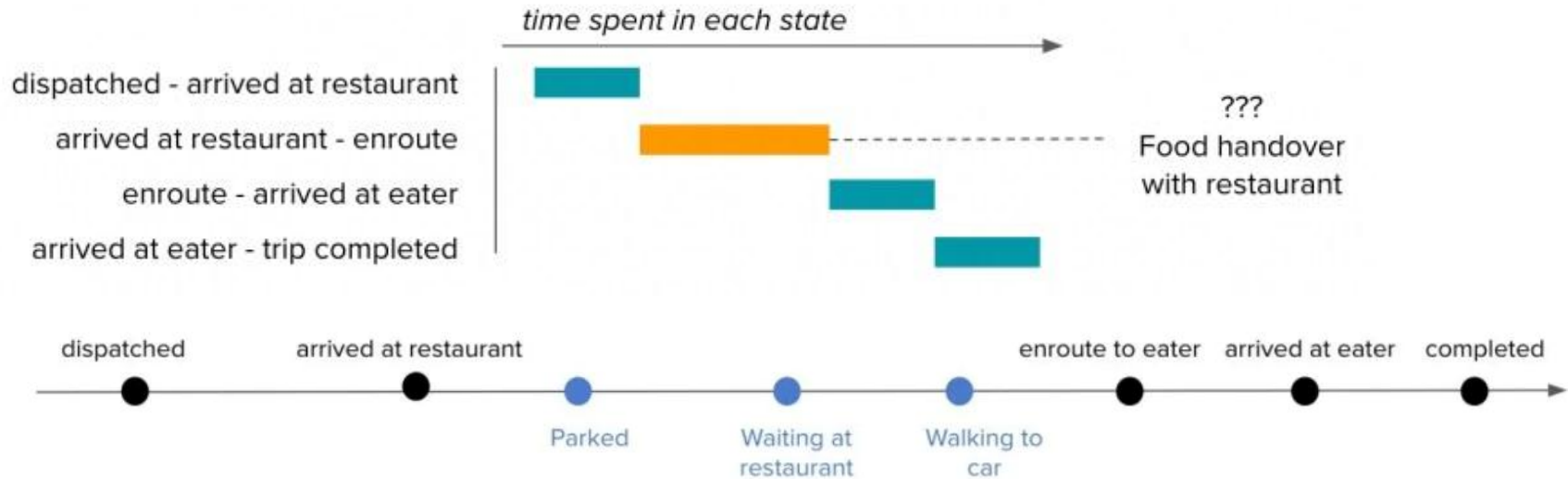
Interaction of partner



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

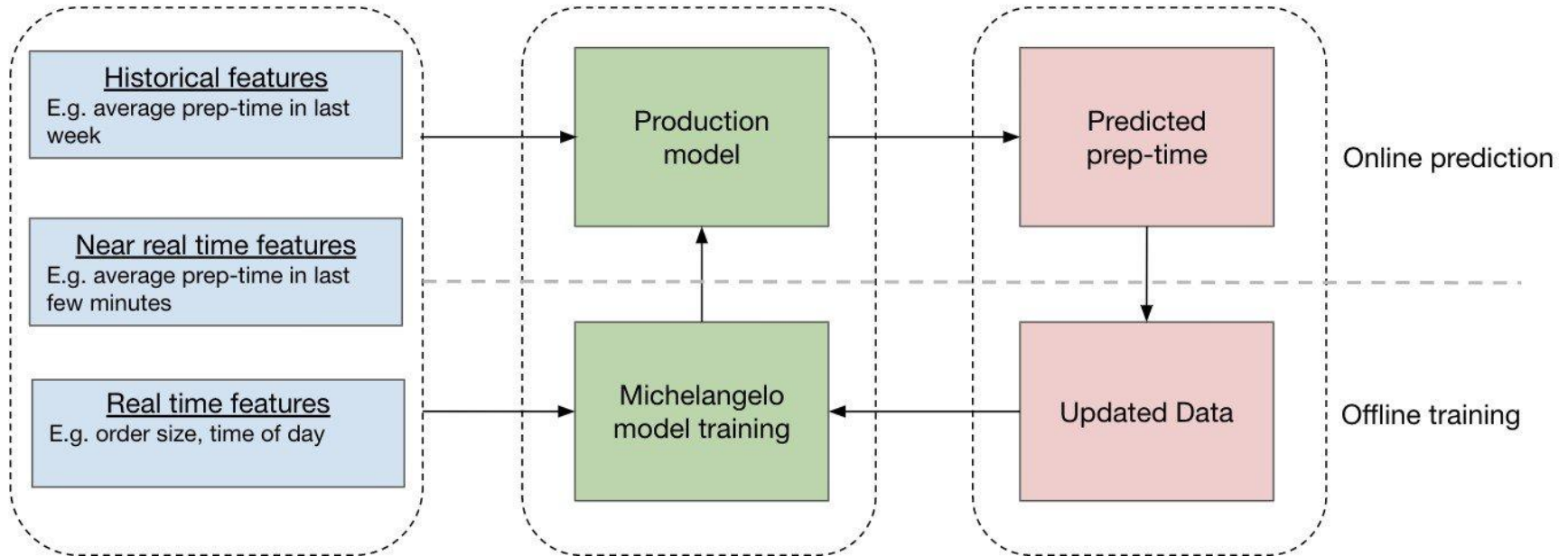
Food preparation at the restaurant

- After order, speedy meal preparation and delivery is of utmost importance
- A crucial point of time is the food handover to the driver that cannot be observed

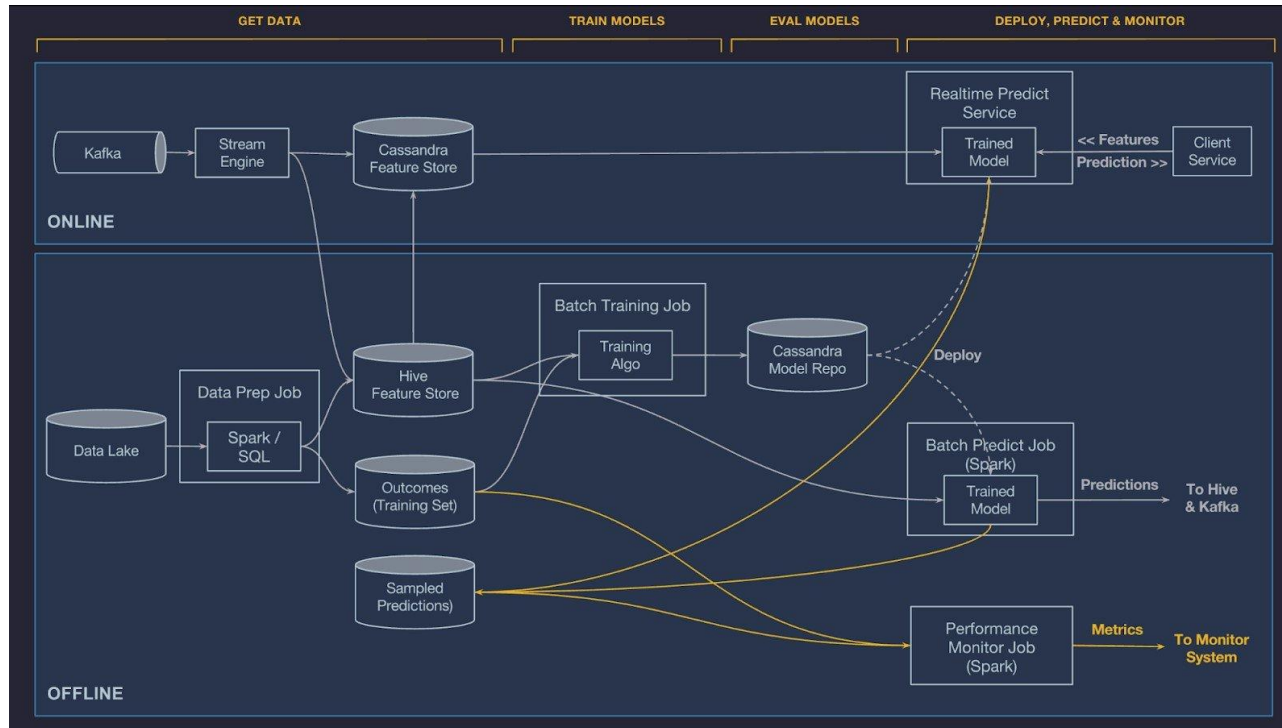


<https://www.uber.com/en-DE/blog/uber-eats-recommending-marketplace/>

Online and offline prediction of food handover time



Data acquisition, training, evaluation and deployment



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Analysis of driver's activities for selected restaurants

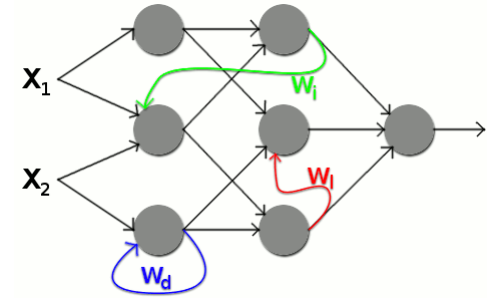
Restaurants are ranked in accordance to duration and variation of food handover time



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Prediction of food handover at restaurant

- Predictive modeling task, supervised learning
- Inputs like restaurant, time of day, #orders placed, size of order, driver, dispatch time
- Predicted output is the time span between driver arrival and handover of food
- Phases of the process depend on each other → Markov model
- Prediction e.g. by recurrent neural networks, that couple ANN neurons in a cyclic way
- Application of RNN e.g. in large language models, where predictions are chained



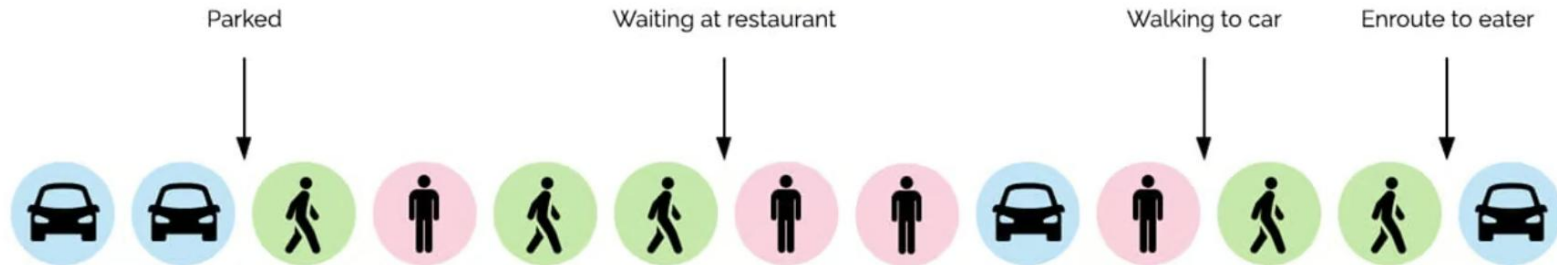
https://de.wikipedia.org/wiki/Rekurrentes_neuronales_Netz#/media/Datei:Neuronal-Networks-Feedback.png

Adjusting prediction by real world observation

Activity modes that are recognized by Android OS

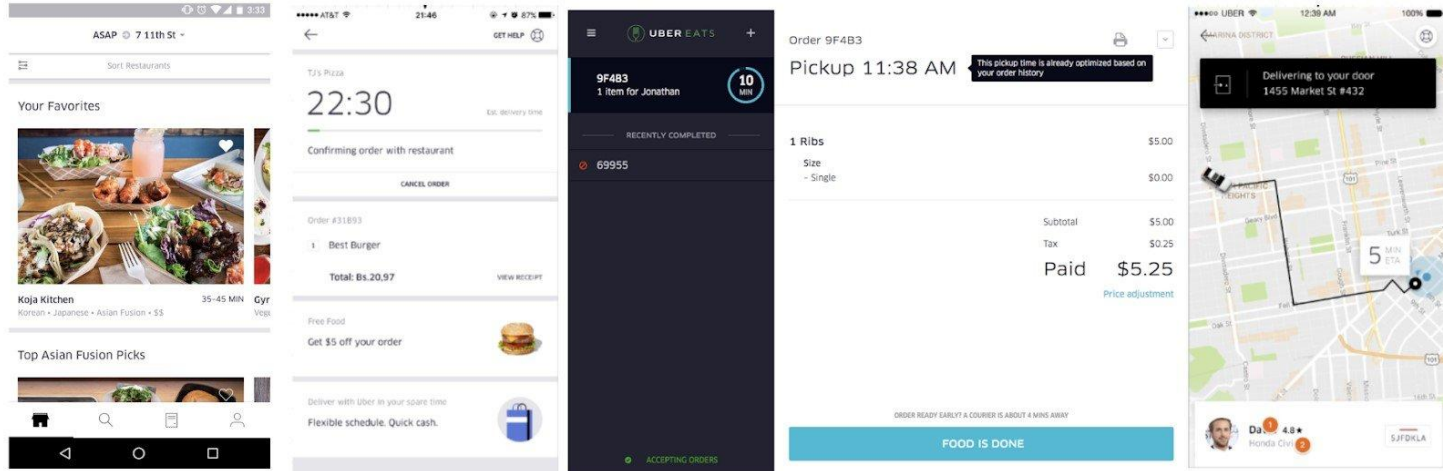


Sequence of activities to be carried out by driver



<https://www.uber.com/en-DE/blog/uber-eats-trip-optimization/>

From order creation to food drop-off



Eyeball ETD prediction

ETD prediction

Prep-time prediction

ETA prediction



<https://www.infoq.com/fr/articles/uber-eats-time-predictions/>

Summary

- Platform business models require online automated decision making
- Decision making incorporates prediction by means of data analytics
- Online optimization ensures seamless transaction of meal delivery
- Expected realization of individual objectives determine food recommendation
- Realized recommendations determine conversion rate → to be maximized

Literature

- Arriving Now: The New Uber Eats (2020)
<https://www.uber.com/newsroom/arriving-now-the-new-uber-eats/>
- Food Discovery with Uber Eats: Recommending for the Marketplace (2018)
<https://www.uber.com/en-DE/blog/uber-eats-recommending-marketplace/>
- Food Discovery with Uber Eats: Building a Query Understanding Engine (2018)
<https://www.uber.com/en-BE/blog/uber-eats-query-understanding/>
- How Does Uber Eats Assign Drivers? (2024)
<https://www.withpara.com/blog/how-does-uber-eats-assign-drivers>
- How Trip Inferences and Machine Learning Optimize Delivery Times on Uber Eats (2018)
<https://www.uber.com/en-DE/blog/uber-eats-trip-optimization/>
- Uber Eats Weather Predictions (2020)
<https://www.infoq.com/fr/articles/uber-eats-time-predictions/> (fr)